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CECS 571: Semantic Web Technologies Project 1 Report

1. Methodology

We evaluated three large language models (LLMs) on their ability to interpret corner-case driving scene images, generate RDF/Turtle semantic descriptions, and identify safety risks with severity scores:

- **Gemini** (Google, gemini-2.5-flash)
- **LLaMA** (Meta, llama-4-scout-17b via Groq API)
- **Claude** (Anthropic, claude-sonnet-4-6) — used as a substitute for ChatGPT due to API availability

Dataset:
We used a dataset of 100 scenarios, comprising 312 resolved image frames, covering six weather conditions (Day/Night × Clear/Foggy/Rain) and three anomaly types (Behavior, Evidence-Based, State). Scenarios without image files were automatically skipped.

Pipeline:
We implemented a processing pipeline (`run_pipeline.py`) that handles images on a per-scenario basis with configurable batch sizes (1, 3, or 5). Each batch is sent as a single API call. The system runs with four concurrent workers and applies exponential backoff with up to five retries to handle transient errors.

Prompt:
We used the standard prompt from section 15 of the assignment verbatim for all models, without any per-model modifications. The prompt asks the models to perform entity identification, relationship identification, generate Turtle RDF triples, identify risk classes, and assign severity scores between 0 and 1.

Ground truth:
We annotated 15 scenarios with three risk labels each, ensuring coverage of all anomaly types and five weather conditions. For evaluation, we used asymmetric token recall: a prediction is considered a true positive if at least 50% of the content tokens from a ground-truth label appear in the predicted text. This approach helps address the mismatch between concise labels and more verbose LLM-generated outputs.

2. Results & Analysis

Table 1 — Risk Discovery

Model	Scenario	Weather	F1	Hall. Rate
Gemini	(avg across 45 GT rows)	All	0.233	0.691
Claude	(avg across 45 GT rows)	All	0.159	0.859
LLaMA (Groq)	(partial: 10/300 calls)	All	0.013	0.079*
Gemini	Unpredictable Taxi	Day-Clear	0.315	0.571

Model	Scenario	Weather	F1	Hall. Rate
Gemini	Bus Obscures Car	Day-Clear	0.364	0.556
Gemini	Fallen Tree	Night-Clear	0.267	0.667
Gemini	Blinking Yellow Int.	Night-Foggy	0.364	0.571
Gemini	Haybale on Highway	Night-Rain	0.364	0.571
Claude	Unpredictable Taxi	Day-Clear	0.129	0.929
Claude	Bus Obscures Car	Day-Clear	0.194	0.893
Claude	Workers Blocked by Van	Day-Clear	0.182	0.900
Claude	Fallen Tree	Night-Clear	0.182	0.900
LLaMA	Unpredictable Taxi	Day-Clear	0.074	0.867
LLaMA	Blinking Yellow Int.	Day-Foggy	0.067	0.857

Table 1. F1 and Hallucination Rate per model and scenario (top 3 rows = per-model averages over 45 GT rows). *LLaMA Hall. Rate based on 10 calls with non-empty output only.

We found that Gemini achieved the highest average F1 score (0.233) and was the only model that consistently identified correct risks across all weather conditions. Claude, on the other hand, tended to over-generate risks—producing around 27 to 35 per scenario. While this gave it a relatively strong recall, it also led to a very high hallucination rate (0.859).

LLaMA's performance was heavily constrained by Groq's free-tier limit of 500K tokens per day. Out of the 522 planned API calls, we were only able to complete 99, and just 10 of those produced non-empty outputs. As a result, its near-zero F1 score reflects this data limitation rather than the model's actual capability.

Table 2 — Performance

Model	Avg Runtime/Batch	Avg Runtime/Scenario	Avg InputSize	Avg OutputSize
Gemini	29.82 s	~90–120 s	2,317,851 B	7,677 chars
Claude	33.94 s	~90–160 s	2,317,857 B	2,701 chars
LLaMA (Groq)	14.16 s	~65–90 s*	1,506,677 B	2,414 chars

Table 2. Runtime and output size (batch size = 1). *LLaMA scenario runtime estimated from bs=1 data only.

We observed that LLaMA running on Groq was the fastest model, averaging about 14 seconds per batch, likely due to LPU hardware acceleration. Gemini, meanwhile, generated the most detailed outputs—about 7,677 characters per batch compared to roughly 2,700 for the other models—indicating richer scene descriptions. Claude was the slowest, taking around 34 seconds per batch. Input sizes were generally similar for Gemini and Claude (about 2.3 MB per batch), while LLaMA processed smaller batches because its quota was exhausted early.

Table 3 — Batch Size Comparison

Model	Batch Size 1	Batch Size 3	Batch Size 5
Gemini	30.27 s	29.86 s	29.33 s
Claude	24.42 s	36.03 s	41.36 s
LLaMA (Groq)	12.21 s	25.56 s*	19.84 s*

Table 3. Avg runtime per batch at batch sizes 1, 3, and 5. *LLaMA bs=3 (n=3) and bs=5 (n=6) are partial.

We observed that Gemini's runtime remained almost constant across different batch sizes (30.27 → 29.33 seconds), suggesting that per-call overhead dominates its latency. This means we can increase batch size to improve throughput without incurring additional latency. In contrast, Claude's runtime increased significantly with larger batches (24 → 41 seconds, +69%), indicating that its generation time scales with token count.

In terms of semantic quality, Gemini maintained consistent performance even as batch size increased. Claude, however, showed no improvement in this regard—its hallucination rate remained consistently high (around 0.86) regardless of batch size.

3. Weather Condition Analysis

Weather	Avg F1	Avg Hall. Rate	Best Model (F1)	Notes
Day-Clear	0.118	0.533	Gemini (0.201)	LLaMA data sparse
Day-Foggy	0.130	0.696	Gemini (0.227)	Highest hallucination rate
Night-Clear	0.177	0.527	Gemini (0.290)	Claude: 0.219
Night-Foggy	0.144	0.562	Gemini (0.324)	Claude: 0.108
Night-Rain	0.221	0.507	Gemini (0.362)	Lowest hallucination rate

Table 4. F1 and Hallucination Rate by weather condition (averaged across all models and GT scenarios).

We found that Gemini performed best in daytime conditions, achieving F1 scores of 0.201 in clear weather and 0.227 in foggy conditions. In contrast, LLaMA performed worst at night, with an F1 score of 0.000 in both Night-Foggy and Night-Rain scenarios, largely due to quota exhaustion. Among the models we were able to fully evaluate, Claude consistently underperformed compared to Gemini at night—for example, in Night-Foggy conditions, it achieved 0.108 versus Gemini's 0.324.

We also observed that foggy conditions led to the highest hallucination rates (Day-Foggy: 0.696), which aligns with the idea that visual ambiguity encourages more speculative risk generation. Overall, Gemini emerged as the most robust model, maintaining the highest F1 scores across all conditions and even showing improved performance in more challenging scenarios like Night-Rain (0.362).

4. Discussion

Discussion Question	Finding
Most accurate semantic output	Gemini — highest F1 (0.233), most detailed outputs (7,677 chars), well-structured Turtle RDF
Most correct risks discovered	Gemini — best true positive count; highest recall across GT scenarios
Most hallucinated risks	Claude — hallucination rate 0.859; generated 25–40 risks/scenario, most unsupported by scene
Batch size effect	Gemini: invariant (efficiency gain at no latency cost). Claude: +69% latency from bs=1 to bs=5.
Most robust across weather	Gemini — highest F1 in every weather condition, smallest variance
Accuracy vs. efficiency	No. LLaMA fastest but near-zero accuracy. Gemini most accurate at ~30 s/batch.
LLM limitations	High hallucination (50–86%), over-generation, no spatial grounding, free-tier quota constraints

Table 5. Summary responses to the seven discussion questions from §16 of the assignment.

LLaMA results are limited due to API quota constraints and should be interpreted with caution.